

SUMMARY

Physics PhD candidate specializing in the intersection of machine learning and quantum information science & computing with 4+ years of research experience.

Research interests: Quantum information science / Quantum computing / Machine learning application

EDUCATION

Aug. 2022 – exp. Aug. 2027	Cornell University PhD candidate in Physics M.S. in Physics, May 2025 Advisor: Prof. Eun-Ah Kim	NY, USA
2018 – 2022	Gwangju Institute of Science & Technology (GIST) B.S. in Physics, minor in Mathematics (Major GPA: 4.47/4.5) - Thesis: Variational Autoencoder for the metal-insulator transition in DMFT - Advisor: Prof. Dong-Hee Kim	Gwangju, S. Korea

RESEARCH EXPERIENCE

2022 – pres.	Kim Research Group, Cornell Physics Graduate Research Assistant (Advisor: Prof. Eun-Ah Kim)	NY, USA
2025 – 2026	Reinforcement learning for Hardware-aware Reconfigurable Atom Array Compilation - Currently developing RL agent inspired by LLM; Large Atom Model (LAM) for reconfigurable atom array compilation which outperforms heuristic and flexible in zoned geometry. - Optimizing qubit movement, gate scheduling, and hardware constraints. Multi-Core sparse-attention Circuit Decoder for Surface code - Currently working on sparse-attention implementation on the modular neural decoder for logical circuit decoding on the surface code, targeting scalable decoder in code distance.	
2024 – 2026	Learning the phases of monitored quantum dynamics - Developed a scalable, data-centric framework for detecting measurement-induced phase transitions (MIPTs) in monitored circuits without post-selection or classical simulation, enabling experimental observability on near-term quantum hardware - Manuscript under review in PRX Intelligence. Experimental collaboration with Quantinuum in the ongoing follow-up work.	
2022 – 2024	Attention to Quantum Complexity - Introduce Quantum Attention Network (QuAN) that leverages the power of attention mechanisms driving large language models (LLM) to characterize complexity of quantum states from bit string data. - Resolved three open problems (i) witnessing entanglement scaling transition, (ii) detecting the growth of quantum complexity in deep random quantum circuit despite the noise, (iii) revealing mixed-state topological order of toric code. - Collaboration with Google Quantum AI and MIT. Published in Science Advances.	

Undergraduate Research Intern (Advisor: Prof. Dong-Hee Kim)

- Variational Autoencoder for the metal-insulator transition in DMFT
- Interpretation of a minimal neural network to learn the metal-insulator transition
- Machine-learning accelerated Monte-Carlo algorithm for 2D spin model

PUBLICATIONS

- [1] **H. Kim**, A. Kumar, Y. Zhou, Y. Xu, R. Vasseur, E.-A. Kim, Learning measurement-induced phase transitions using attention, *arXiv:2508.15895* (under review in PRX Intelligence, 2026).
- [2] J. Tahmassebpur, A. Bhuiyan, **H. Kim**, O. Lesser, Learning from almost nothing: How neural networks survive heavy input corruption, *arXiv:2606.11319*.
- [3] **H. Kim**, Y. Zhou, Y. Xu, K. Varma, A. H. Karamlou, I. T. Rosen, J. C. Hoke, C. Wan, J. P. Zhou, W. D. Oliver, Y. D. Lensky, K. Q. Weinberger, E.-A. Kim, Attention to Quantum Complexity, *Sci. Adv.* **11**, eadu0059 (2025).
- [4] **H. Kim**, D. Kim, D.-H. Kim, Minimal Neural Network to Learn the Metal-insulator Transition in the Dynamical Mean-field Theory, *NPSM 2022*, **72**, 487-494 (2022).
- [5] **H. Kim**, Y. Xu, J. P. Zhou, E.-A. Kim, Reinforcement learning for Hardware-Aware Reconfigurable Atom Array Compilation (in preparation, 2026)

PRESENTATIONS

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| Invited talks | <ul style="list-style-type: none"> [1] H. Kim. AI for Quantum: Decoding logical circuit and Compilation on Neutral atom platform, <i>Research Seminar, LG Electronics</i>, Seoul, S. Korea, Jun. 2026. [2] H. Kim. Reinforcement learning for hardware-aware reconfigurable atom array compilation, <i>Lunch Seminar, Department of Quantum Information Engineering, Sungkyunkwan University</i>, Suwon, S. Korea, Jun. 2026. [3] H. Kim. Attention to Quantum Complexity: Learning measurement-induced phase transitions, <i>Group Seminar, Department of Physics, KAIST</i>, Daejeon, S. Korea, Jun. 2026. [4] H. Kim. Learning Measurement-Induced Phase Transition Using Attention, <i>QST Seminar, Seoul National University</i>, Seoul, S. Korea, Jun. 2026. [5] H. Kim. Attention to Quantum Complexity: Learning measurement-induced phase transitions, <i>Department of Physics, Sejong University</i>, Seoul, S. Korea, Jun. 2026. [6] H. Kim. Learning Measurement-Induced Phase Transition Using Attention, <i>Seminar, Korea Institute for Advanced Study</i>, Seoul, S. Korea, Jun. 2026. [7] H. Kim. Reinforcement Learning for Hardware-Aware Reconfigurable Atom Array Compilation, <i>AI and Natural Sciences Seminar, Korea Institute for Advanced Study</i>, Seoul, S. Korea, Jun. 2026. [8] H. Kim. Learning Measurement-Induced Phase Transition Using Attention, <i>Special Seminar, GIST</i>, Gwangju, S. Korea, Jun. 2026. [9] H. Kim. Learning Measurement-Induced Phase Transition Using Attention, <i>Seminar, Yonsei University</i>, Incheon, S. Korea, Jun. 2026. |
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Contributed talks

- [10] **H. Kim**, Attention to quantum complexity: learning phases of monitored quantum dynamics, In *APS Joint March and April Meeting Global Physics Summit*, 2025
- [11] **H. Kim**, Y. Xu, J. P. Zhou, K. Q. Weinberger, E.-A. Kim. Reinforcement Learning for Hardware-Aware Reconfigurable Atom Array Compilation, In *APS Global Physics Summit*, 2026
- [12] **H. Kim**, A. Kumar, Y. Zhou, Y. Xu, R. Vasseur, E.-A. Kim. Learning the phases of monitored quantum dynamics - II, In *APS Joint March and April Meeting Global Physics Summit*, 2025
- [13] **H. Kim**, Y. Zhou, Y. Xu, C. Wan, J. P. Zhou, Y. Lensky, J. Hoke, P. Roushan, K. Weinberger, E.-A. Kim. Attention to complexity III: learning the complexity of random quantum circuit states, In *American Physical Society March Meeting*, 2024
- [14] **H. Kim**, K. Varma, C. Wan, Y. Zhou, Y. Lensky, K. Weinberger, E.-A. Kim. Detecting quantum complexity using transformer-based neural network (II), In *American Physical Society March Meeting*, 2023
- [15] **H. Kim**. Attention to Quantum Complexity, In *Cornell Quantum Day Fall 2023*
- [16] **H. Kim**, D. Kim, D.-H. Kim. Interpretation of a minimal neural network to learn the metal-insulator transition in the dynamical mean-field theory, In *2021 Korean Physical Society Conference in Gwangju-Chonnam Branch*, 2021

SKILLS

Programming Languages

- Language: Python, C/C++
- ML framework: PyTorch, gymnasium
- Quantum simulation: Qiskit, Cirq, Pytket

ACADEMIC HONORS

2020 – 2021	National Scholarship for Science & Engineering , Korea Student Aid Foundation, full fund
2018 – 2019	Korean Government Scholarship , GIST
2018 – 2020	Academic Scholarship to excellence , GIST

TEACHING EXPERIENCE

	Cornell University	<i>NY, USA</i>
	◦ Teaching assistant	
<i>Fall 2025</i>	... PHYS 2214: Oscillations, Waves, and Quantum Physics: Manage discussion sections, lab sections, homework, and exams.	
<i>Fall 2022</i>	... PHYS 2213: Electromagnetism: Manage discussion sections, homework, and exams.	